

Designator	Component	Value / Part Number
U2	ESP32 Module	ESP32-WROOM-32
S1	Temperature & Humidity Sensor	AM2302 / DHT22
U1	Voltage Regulator	NCP1117LPST33T3G
U3	USB to UART IC	CP2102N-A02-GQFN28R
U4	USB Type-C Connector	USB Type-C
R1	Resistor	5.1k Ω
R2	Resistor	560 Ω
R3	Resistor	270 Ω
R4	Resistor	270 Ω
R5	Resistor	10k Ω
R6	Resistor	10k Ω
R7	Resistor	22.1k Ω
R8	Resistor	47.5k Ω
R9	Resistor	270 Ω
R10	Resistor	270 Ω
R11	Resistor	5.1k Ω
R12	Resistor	5.1k Ω
R13	Resistor	1k Ω
R14	Resistor	10k Ω
R15	Resistor	10k Ω
R16	Resistor	270 Ω
R17	Resistor	270 Ω
R18	Resistor	10k Ω
R19	Resistor	10k Ω
C1	Capacitor	100nF
C2	Capacitor	100nF
C3	Capacitor	22nF
C4	Capacitor	100nF
C5	Capacitor	100nF
C6	Capacitor	4.7 μ F
C7	Capacitor	100nF
C8	Capacitor	100nF
C10	Capacitor	100nF
C11	Capacitor	100nF
C12	Capacitor	100nF
C13	Capacitor	100nF
D1	LED	Red LED
D2	LED	Red LED
U5-U7	ESD Protection Diodes	TVS Diode
Q1	Transistor	N-MOSFET
Q2	Transistor	N-MOSFET
Q3	Transistor	NPN Transistor
Q4	Transistor	NPN Transistor
SW1	Push Button	Reset Button
SW2	Push Button	Boot Button
J1	Jumper/Header	3-Pin Header
H1	Header Connector	16-Pin Header

H2

Header Connector

16-Pin Header

Quantity	Description
1	Main microcontroller module
1	Digital temperature and humidity sensor
1	5V to 3.3V voltage regulator
1	USB-to-Serial converter
1	USB communication and power connector
1	Power LED current limiting resistor
1	User LED current limiting resistor
1	IO protection resistor
1	IO protection resistor
1	USB pull-up resistor
1	Reset pull-down resistor
1	USB/UART configuration resistor
1	USB/UART configuration resistor
1	USB D+ line resistor
1	USB D- line resistor
1	USB Type-C configuration resistor
1	USB Type-C configuration resistor
1	GPIO resistor
1	GPIO pull-down resistor
1	EN pull-up resistor
1	RXD serial resistor
1	TXD serial resistor
1	DTR transistor control resistor
1	RTS signal transistor control resistor
1	Input decoupling capacitor
1	Output decoupling capacitor
1	ESP32 filtering capacitor
1	ESP32 decoupling capacitor
1	Connector filtering capacitor
1	USB-UART stabilization capacitor
1	Connector filtering capacitor
1	Power filtering capacitor
1	USB Type-C filtering capacitor
1	GPIO filtering capacitor
1	Reset circuit capacitor
1	Boot circuit capacitor
1	Power indicator LED
1	User status LED
3	USB ESD protection
1	Auto-reset control transistor
1	Auto-boot control transistor
1	DTR signal transistor
1	RTS signal transistor
1	ESP32 reset switch
1	ESP32 boot mode switch
1	Power source selection jumper
1	ESP32 GPIO breakout connector

1 ESP32 GPIO breakout connector